

Learning track: building ppa-lab like a quant on a commercial team

This folder is your study guide for the project. Each day has one page that explains **what was built, why it was built that way, the market and maths behind it, and how the same work is done inside a company like Encavis**. PDF versions are in [pdf/](#) (rebuild them with `make docs`).

How to use a day's page

Work through each page in four passes. That is how the material sticks, and how you become able to explain it in an interview.

1. **Read** the concepts section with a notebook next to you (about 60-90 minutes).
2. **Run** the hands-on commands and check that you get the same results.
3. **Change** something: do the exercises; break a test on purpose and fix it.
4. **Explain**: answer the quiz without looking, then say the 60-second interview story out loud, in English and once in German.

Tick the checklist at the end of each page. Do not move to the next day until you can explain every box in your own words.

Pages

Day	Ticket	Page	Topic
1	PPA-1	day01_market_data_foundation.md	Power-market basics, time and units, a real data pipeline, data quality, reconciliation
2	PPA-2	(next)	Capture prices and capture rates: what solar and wind really earn

Conventions used in every page

- **Real-world note**: how the same task looks inside a utility or IPP.
- **Pitfall**: a mistake that costs real money if it reaches a pricing model.
- **Interview**: phrasing you can reuse when talking about this work.
- Code references point to files in this repository, e.g. `src/ppa_lab/timeutils.py`.